

Public Comment Invited on Artificial Intelligence Action Plan

Prepared by The Chemours Company

The Chemours Company appreciates the opportunity to provide input to the Office of Science and Technology Policy (OSTP) regarding the Artificial Intelligence (AI) Action Plan. Chemours is a global leader in Thermal & Specialized Solutions (TSS), Titanium Technologies (TT), and Advanced Performance Materials (APM), providing its customers with solutions in a wide range of industries with market-defining products, application expertise and chemistry-based innovations.

AI will enable the U.S. to accelerate innovation, and therefore competitiveness, across key industries such as manufacturing, financial services, technology, and defense. As an active user of AI in our daily operations and innovation processes, Chemours recognizes AI's critical role in maintaining competitive advantage across industries. We also recognize that as data centers evolve to meet the demands of the AI boom, they are generating more heat that must be managed.

As a global leader in advanced cooling solutions, our TSS business develops innovative technologies that support critical infrastructures like data centers, which are foundational to AI advancements, as well as solutions for the HVACR, automotive, and semiconductor industries. Our portfolio includes industry-leading products such as Freon™ 410A, which plays a key role in the thermal management of data centers via traditional air-cooling methods. We also have an advanced two-phase immersion cooling (2-PIC) technology, [Opteon™ 2P50](#), in development. When using immersion cooling, all IT hardware (servers, chips, etc.) are placed inside a sealed tank and submerged in a dielectric fluid, which enables efficient heat removal without interfering with the functionality or safety of the electrical components. This deep expertise in cooling technologies positions Chemours as a trusted partner in enabling the advanced computing and AI ecosystems that are transforming the world.

As AI computing demands increase, so do the challenges of managing energy, water, and space constraints. Chemours' advanced 2-PIC technology provides a solution to these challenges, offering superior heat management for chips and server components. This capability will ensure scalability, reliability, and peak performance for AI infrastructure critical to U.S. innovation and national security.

By enabling high-performance AI systems and infrastructure to be scaled in an energy and water efficient manner, 2-PIC aligns with the priorities of the current administration to grow the U.S.'s AI leadership, re-establish economic competitiveness, and reduce reliance on resource-intensive methods.

In addition to immersion cooling technologies, Chemours' APM business offers specialty fluoropolymers and technologies to enable the high-speed connectivity within data center servers, through multiple wire and cable solutions. Such fluoropolymers provide critical insulating properties to ensure high-speed data communications within and across AI computing servers. Without Chemours' fluoropolymer solutions, the computing power of AI data centers would be jeopardized.

To achieve the Administration's goal of dominating the global AI race, the U.S. must reduce regulatory bottlenecks and improve efficiency, transparency and communication to ensure new innovations, such as 2-PIC, can help solve the challenges surrounding AI growth in America.

AI Growth Drives Data Center Expansion

AI technology is transforming industries worldwide, fueling exponential growth in data processing and storage as well as significant investment from enterprises as they grapple with the introduction of these tools. As AI models become more sophisticated, the demand for high-performance computing systems intensifies. Traditional data center cooling technologies are simply inadequate for the U.S.'s AI leadership aspirations. They are inefficient due to their high energy and water consumption coupled with a lack of heat transfer capacity to support future generations of IT components. Data centers, which are vital for AI computing, consumed about 4.4% of total U.S. electricity in 2023 and are expected to consume approximately 6.7 to 12% of total U.S. electricity by 2028, according to the Department of Energy¹. As of 2023, it is estimated that data centers account for more than 17.4 billion gallons of direct water consumption annually, while the indirect water consumption was estimated to be over 211.3 billion gallons mainly driven by power generation². These alarming values are only expected to increase without a substantial change in technology.

Within AI data centers, there are hundreds of miles of data transmission cables that connect the chips and AI servers. Data needs to be exchanged at very high speed through these cables with extremely low delays in a very confined space within AI servers. Non-fluorinated polymers in such cables are challenged to achieve these requirements for AI servers and currently cannot meet performance requirements for the most advanced AI server applications.

The Need for Next-Generation Cooling Solutions

Cooling is traditionally underrepresented in conversations about AI and data centers; however, cooling architectures have a significant impact on overall data center design, power use, and water consumption. Traditional cooling approaches are reaching their performance limits in the face of skyrocketing thermal demands. For instance, over 40% of data center energy consumption is dedicated solely to cooling IT equipment³, and with power requirements increasing for next generation chips, a logical area to explore without limiting IT equipment development is improving and optimizing cooling systems.

In April 2024, Goldman Sachs reported that energy efficiency improvements have fallen to levels not experienced in well over a decade. Prior to 2021, efficiency improvements were driving double digit year over year improvements in data centers, however since 2022 energy efficiency improvements have decreased to less than 4% and are expected to hold here through 2030 without substantial changes in the technologies being deployed⁴.

Next-generation GPUs and CPUs, which form the backbone of AI systems, are generating unprecedented amounts of heat, far surpassing the capabilities of traditional air-cooling

¹ <https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers>

² <https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf>

³ <https://www.nrel.gov/news/program/2023/nrel-joins-effort-to-advance-data-center-cooling-efficiency.html>

⁴ <https://www.goldmansachs.com/images/migrated/insights/pages/gs-research/gs-sustain-generational-growth-ai-data-centers-global-power-surge-and-the-sustainability-impact/sustain-data-center-redaction.pdf>

systems and soon, alternative liquid-cooling architectures. This rapid escalation in thermal output is not just a technical challenge but a critical bottleneck threatening the performance and reliability of AI infrastructure. Without innovative cooling solutions, the deployment of AI technologies could face significant obstacles, limiting their scalability and effectiveness.

2-PIC technology from Chemours offers a transformative solution to these issues, providing unparalleled efficiency in heat transfer while dramatically reducing energy consumption, water usage, and spatial demands compared to traditional air- and liquid-cooled systems. By enabling the reliable operation of next-generation AI GPUs and CPUs, 2-PIC technology not only eliminates bottlenecks in performance but also supports the sustainable growth of AI infrastructure, ensuring its viability for future applications.

A Transformative Solution for AI Infrastructure: The Key Benefits of 2-PIC Technology

1. Energy Efficiency:

2-PIC reduces cooling energy consumption by up to 90% when compared to air-cooled systems, eliminating the need for pumps, fans, and air conditioning. If deployed globally, this could generate 340 TWh in energy savings by 2055, equivalent to powering more than 31 million households annually.

2. Water Conservation:

Unlike traditional cooling methods, which can consume millions of gallons of water annually at a single site⁵, 2-PIC operates with nearly zero water use in most climates. This significantly reduces strain on local water resources, particularly in drought-prone areas.

3. Compact Design:

2-PIC reduces the physical footprint of cooling infrastructure by up to 60%, when compared to air-cooled systems, which require adequate spacing between server racks. This allows data centers to optimize their space for increased computing capacity while lowering capital and operational expenditures.

4. Improved Equipment Performance:

Opteon™ 2P50 operates at 50°C (122°F) consistently enabling systems to maintain an optimum temperature for the IT components while also providing sufficient heat that can be extracted from the fluid for use in other applications. The primary environmental factors that cause IT components to fail include uneven temperatures across components, overheating, accumulation of dust, and high humidity. When leveraging 2-PIC, your IT equipment is operating in a sealed, dust-free environment with consistent temperatures and no humidity, effectively extending the lifespan of IT components, which lowers maintenance costs and reduces electronic waste.

5. Heat Transfer Capability and Capacity:

Two phase fluids in general have a higher heat transfer potential than single phase fluids in use today such as water. This capacity is based on the phase change between a liquid and a vapor which unlocks the fluids' ability to utilize latent heat of vaporization. This enables two phase fluids to have 10-100 times greater capacity than single phase alternatives ensuring investments in the technology are usable for years to come.

⁵ <https://www.weforum.org/stories/2024/11/circular-water-solutions-sustainable-data-centres/>

6. Comparative Baseline Study on Performance and Total Cost

Compared to leading alternative solutions, namely single-phase immersion cooling and single phase direct to chip cooling, 2-PIC significantly outperforms the alternatives on a performance and cost basis. To validate the capabilities, value, and economic feasibility of deploying 2-PIC in data centers, we modeled a 36 MW data center build utilizing Chemours 2-PIC technology, Opteon™ 2P50, in Leesburg, Virginia, which demonstrated:

- Average Capital Cost Reduction of 6%
- Average Operational Cost Reduction of 9%
- Average Reduction in PUE (a measurement of power consumption) of 8%
- Average reduction of the total cost of ownership of 8%

7. Environmental Stewardship:

Chemours' Opteon™ 2P50 dielectric fluid, a hydrofluoroolefin (HFO) based chemistry, combines ultra-low global warming potential (GWP) with circularity capabilities, allowing for reprocessing and reuse to minimize environmental impact.

Chemours recognizes the importance of cross-sector collaboration to accelerate the deployment of advanced technologies like 2-PIC and fluoropolymers in high-speed AI data center cables. We encourage OSTP to include the following in the AI Action Plan:

1. Support for Advanced Cooling Solutions:

We recommend the Trump Administration provide government support to further research through relevant federal agencies and encourage collaboration between public and private sectors to support the development and adoption of high-efficiency cooling technologies that align with AI infrastructure needs. Many of the new cooling solutions coming to market require a change in design as well as standard operating procedures. Coalitions with a combination of public and private sector resources can effectively overcome these changes to set the standard for industry.

2. Regulatory Certainty and Overcoming Regulatory Hurdles:

The unique properties of some fluorinated chemistries make them ideal to tackle the challenges in powering and operating data centers and related facilities, and these innovative new molecules can be advanced while still protecting human health and the environment. The current process for bringing a new chemical to market through the EPA Toxic Substances Control Act (TSCA) has been challenged since the law was amended in 2016. The process is arduous and time-consuming, which means innovations often lose their competitive edge in the American market. The agency has an opportunity to improve the process by making it more efficient, transparent, and collaborative with industry to create impactful changes that are durable enough to withstand future attempts at reversals. We encourage EPA to collaborate with industry, use measured data, better define reasonably foreseeable conditions of use, and negotiate consent orders that don't extinguish the market case. These regulatory hurdles have created a bottleneck at EPA and threaten the ability for America to swiftly adopt new technologies that can drive our competitiveness globally.

At Chemours, we believe that AI innovation should be supported by sustainable infrastructure that balances performance, cost, and environmental impact while retaining America's AI leadership position. We appreciate the work the Trump Administration is undertaking to become the global leader in AI and the support we are seeing across the federal government to

ensure we right-size the infrastructure to support these initiatives. We look forward to continued engagement with OSTP and the White House as they develop an AI Action Plan that underscores America's leadership in both AI and innovation.